

Lab Task 9: Building POS Tagger for Unstructured Web Documents

Aim:

To build a POS tagger for unstructured text and extract useful information using PyTorch / Keras.

Code:

```
sentences = [["I", "love", "NLP"], ["NLP", "is", "fun"]]
tags = [["PRP", "VBP", "NN"], ["NN", "VBZ", "JJ"]]
word_tag_map = {}
for sent, tag_seq in zip(sentences, tags):
    for word, tag in zip(sent, tag_seq):
        word_tag_map[word] = tag
test_sentence = ["I", "love", "NLP"]
# Prediction
predicted = [(word, word_tag_map.get(word, "NN")) for word in test_sentence]
print("Predicted POS Tags:")
print(predicted)
print("\nNouns in sentence:")
for word, tag in predicted:
    if tag == "NN":
        print(word)
```

output:

```
Predicted POS Tags:
[('I', 'PRP'), ('love', 'VBP'), ('NLP', 'NN')]
Nouns in sentence:
NLP
```